

Abstraction

Worksheet for the 360° experience: Abstraction studio

Name: _____ Class: _____ Date: _____

How to use this sheet

1. Discuss or research the Do Now on slide 1. Complete the starter and key terminology below.
2. Open Abstraction studio. Read each station panel and write the key facts in your own words. Complete its worksheet tasks before tapping the badge.
3. Answer the questions on screen. Your first answer counts. Use review mode to learn from mistakes.
4. Copy your score into the yellow box, then complete the exam practice from memory.

By the end of the lesson I will:

- Explain abstraction.
- Select details for a stated purpose.
- Justify why some details can be ignored.

Starter: before you put the headset on

A pupil needs to find the library. Which is more useful: a map of corridors or a list of wall colours? Explain.

Key terminology

Abstraction

Model

The stations

Stations 1 and 2: turn right. Stations 3 and 4: turn around. Stations 5 and 6: turn left. Final challenge: look down.

1 Keep the purpose

Turn right

Key facts in your own words:

Challenge: Why can a map omit carpet patterns?

Which detail belongs on a route map?

For recognising a camera icon, which detail is least necessary?

2 Essential information

Turn right

Key facts in your own words:

Challenge: Name two details needed to track a camera loan.

Classify details for tracking loans.

Categories: Needed / Not needed

Item	Category
Borrower ID	
Item ID	
Favourite film	
Shoe brand	

Which detail helps identify an overdue loan?

3 Remove distractions

Turn around

Key facts in your own words:

Challenge: Why might an icon omit the serial number?

What would damage a route map?

Removing required corridor links is...

4 Purpose changes relevance

Turn around

Key facts in your own words:

Challenge: When would a room light fault become relevant?

Match purpose to a useful detail.

Match each item to a description from the list below.

Borrower ID / Fault description / Corridor connections

Item	Matching description
Find a room	
Repair a camera	
Return a loan	

A fault description becomes relevant when...

5 Abstraction in software

Turn left

Key facts in your own words:

Challenge: What does a return button hide from its user?

A return button is an abstraction because it...

Hiding processing behind a button means...

6 Evaluate a model

Turn left

Key facts in your own words:

Challenge: How should this route map be improved?

Which justification is strongest?

A model should be judged against...

★ Final challenge (look down)

Before you tap the star, complete these tasks on paper.

Choose data for identifying overdue equipment loans.

Categories: Keep / Omit

Item	Category
Borrower ID	
Due date	
Item ID	
Favourite snack	

Why is abstraction useful here?

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Your score

Copy it from the score screen as soon as it appears.

_____ / 22

Key question

Would a photograph always be more useful than a simple map?

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Review and improve

Use review mode. Explain one correction and why the correct answer fits.

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Exam practice

Near the end of the lesson, close the experience and answer from memory.

a) Define abstraction. [2]

b) A delivery map omits the colour of each house. Explain why this is appropriate. [2]

c) Give one detail that a loan system should retain and justify it. [2]

How confident am I?

Objective	Not yet	Getting there	Confident
Explain abstraction.			
Select details for a stated purpose.			
Justify why some details can be ignored.			